

Fahad Siddique Faisal

SOFTWARE ENGINEER · AI RESEARCHER

Dhaka, Bangladesh

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Research Interest

LLMs, Cybersecurity, IoT, Autonomous Systems, Distributed Learning, ML, Deep Learning, Edge-AI

Work Experience

Walton Hi-Tech Industries PLC (R&D)

AI RESEARCHER & SENIOR DEPUTY ASSISTANT DIRECTOR

Dhaka, Bangladesh

Apr. 2025 - Present

Project: Walton-AI: Local conversational AI and Cosine Similarity based Knowledge Management System

- Developed a fully local AI-based knowledge-base and chat system using open-source AI models and **LangChain** and **Lang-Graph** with a Corrective-RAG pipeline, trained on internal data and deployed on Walton's In-house Ubuntu VPS.
- Developed an AI-based Real-time on-device Speech Recognition and Transcription system utilizing whisper.cpp and open-source LLM models.
- Implemented distributed and Parallel GPU training and inference capability of NVIDIA GPU with **PyTorch**, **CUDA** programming and llama.cpp
- Fine-tuned LoRA Adapters, developed a GPT-2-based Small language model with 20M and 30M parameters with function-calling capability.

Project: AI Long-Term Memory System

- Designed and deployed a persistent memory framework for enterprise AI agents using Agentic Routing (PageIndex), Dynamic Importance Decay and Automated Memory Merging.
- Integrated the memory system with production RAG services and vector databases, improving retrieval efficiency.

Project: Smart Scan

- Collaborated in the development and production deployment of **Smart Scan**, a web-based platform providing refrigerator information, user manuals, product configuration, AI-powered service assistance and issue reporting.
- deployed backend services on a Hostinger Cloud VPS running **Ubuntu Linux**, ensuring secure and scalable application hosting.

AI Solutions USA

SOFTWARE ENGINEER(AI/ML)

New York, USA (Remote)

Jun. 2025 - Jun 2026

Project: PetitionPro – AI-powered NIW Immigration Petition Platform

- Designed a multi-agent orchestration system with **CrewAI**, **Instruct** and **Agno** to orchestrate AI agents for collaborative task execution for a complete NIW petition drafting, personalized Recommendation Letter drafting and CV evaluation.
- Developed AI-powered document processing and Retrieval-Augmented Generation (RAG) services using Gemini and OpenAI API, Pinecone databases and REST APIs with FastAPI for PetitionPro

Project: DroneSeg

- A full-stack semantic segmentation platform for drone & aerial imagery that runs a SegFormer-B2 deep learning model to produce pixel-accurate land-cover classification masks.

Undergraduate Thesis

Investigation and Transforming Resource-Scarce Language Question Generation: A Study on Text-to-Text Transformer Models for Bengali

Bangladesh

AUTHOR

2024

- Developed and improved an automatic FAQ generation system using text-to-text transformer models to enhance Question Generation (QG) in multilingual contexts.

Honors & Awards

INTERNATIONAL

- 2024 **Best Technical Paper in a Session on AI**, 27th International Conference ICCIT2024
- 2022 **Winner**, scholarship for CSCA (Certified Supply Chain Analyst)
- 2022 **Winner**, ISCEA Prize Global Supply Chain

Cox's Bazar

Ohio, U.S.A

Ohio, U.S.A

DOMESTIC

- 2025 **Champion among 170 Engineering Team**, Walton AI Creative Award 2025
- 2024 **Winner**, University Semester Scholarship for Outstanding Result
- 2020 **Winner**, University Semester Scholarship for Outstanding Result
- 2019 **Winner**, University Semester Scholarship for Outstanding Result

Bangladesh

Bangladesh

Bangladesh

Bangladesh

Publications

- **F. S. Faisal**, S. N. Islam, M. F. Hossain and M. A. Hossain. "Optimizing Resource-Scarce Language Question Generation on Bengali: A Comparative Study of Transformer-Based Models," *27th International Conference on Computer and Information Technology (IEEE ICCIT)*, 2024. [\[Paper\]](#)
- S. N. Islam, **F. S. Faisal**, M. F. Hossain and M. A. Hossain, "Enhancing Bilingual Sarcasm Detection with an Advanced Ensemble Model: A Dual-Language Deep Learning Framework," *27th International Conference on Computer and Information Technology (IEEE ICCIT)*, 2024. [\[Paper\]](#)
- M. S. H. Sarker, **F. S. Faisal**, A. Istiaque and T. Taher, "Design and Comparative Analysis of 1-BIT ALU and Full Adder using MGD1 Technique for High-Performance PLC Applications," *27th International Conference on Computer and Information Technology (IEEE ICCIT)*, 2024. [\[Paper\]](#)
- Das, G. Ali, A. Hossain, **F. S. Faisal**. "A Federated and Explainable Machine Learning Framework for Robust Intrusion Detection and Network Security Enhancement in Industrial Internet of Things (IIoT) Environments," *International Conference on Electrical, Computer & Telecommunication Engineering (IEEE ICECTE)*, 2026. [\[Paper\]](#)
- Mithila Arman, Reduanul Bari Shovon, Sonet Barua Emon, Jannatul Asma Sameha, **Fahad Siddique Faisal**, Moshir Rahman Sayem, "Automated Chest X-Ray Report Generation Using Vision-Language Models: A LLaMa 3.2 Based Approach", *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, 2026. [\[Paper\]](#)

Projects

Sensor-Aware Adaptive Linear Trajectory Model for Optimal Sensor Placement

[\[Code\]](#)

KEY WORDS : BARRIER COVERAGE, TRAJECTORY MODELING, SENSOR NETWORKS, PYTHON

2026

- A sensor-aware adaptive extension of LGCP-based barrier coverage that reweights trajectory intensity toward low-detection paths conditioned on sensor deployment, enabling realistic modeling of evasive adversarial targets (drones, UAVs).

AIOT-Based-Smart-Fridge [\[Code\]](#)

TECH STACK: RASPBERRY PI 4B X3, TEMPERATURE AND HUMIDITY SENSORS, YOLO, PYTHON

2026

- Smart Fridge monitors your fridge about temperature and humidity, alerts you when food is about to expire, or when someone steals your food. In addition, it can generate a recipe based on items in the fridge with the help of OpenAI API

Voice Command Fingerprinting - Attack and Defense [\[Code\]](#)

TECH STACK: WIRESHARK, BUFL0, SCIKIT-LEARN

2026

- Implemented the VCFP (Voice Command Fingerprint) attack and BuFLO defense to analyze voice assistant network traffic, showing how encrypted packets can reveal voice commands and how BuFLO mitigates this traffic analysis.

MedConvFormer : Hybrid CNN-Transformer to Distinguish COVID-19, Normal, and Pneumonia from chest X-ray images [\[Code\]](#)

TECH STACK: PYTHON, EFFICIENTNET-Bo, VISION TRANSFORMER (ViT)

2025

- Implemented a novel hybrid deep learning architecture that uniquely combines the strengths of Convolutional Neural Networks (CNNs) and Vision Transformers (ViT) for medical image classification.

AI Long-Term Memory (ALTM) [\[Code\]](#)

TECH STACK: PYTHON, MONGODB ATLAS, PAGEINDEX

2025

- ALTM has Agentic Routing (PageIndex), Dynamic Importance Decay and Automated Memory Merging to ensure agents remember what's important and forget what doesn't.

CLHF-Med-AI: Medical Conversational AI Model Development with Continuous Learning (CLHF) [\[Code\]](#)

TECH STACK: QWEN2, QLoRA

2025

- High-performance medical assistant based on the Qwen2-0.5B-Instruct architecture optimized via 4-bit QLoRA and KTO for Continuous Learning that refines model behavior through real-time human preference data.

Real-Time Image Pattern Detection and Recognition using YOLO, OpenCV and CLIP Embeddings [\[Code\]](#)

TECH STACK: OPENCV, CLIP EMBEDDINGS, YOLOv8, QDRANTDB

2025

- Developed an end-to-end computer vision system combining YOLOv8 for real-time refrigerator detection, CLIP embeddings for feature extraction, and Qdrant vector database for similarity search.

Microcontroller and IoT Projects

TECH STACK: MOSQUITTO MQTT, ESP32, ARDUINO, WHISPER.CPP, AC RELAY & WIFI MODULE.

2022

- Developed an AI-based-voice controlled over-the-internet Automatic Warehouse Switching and Monitoring system.
- Real-time Weather Monitoring and Forecasting system

Education

Chittagong University of Engineering and Technology

Chittagong, Bangladesh

B.SC. IN ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Feb. 2019 - Jul. 2024

- Obtained CGPA of 3.44 out of 4.0 with last semester GPA 3.73

Relevant Coursework

- **Programming Related:** Fundamentals of computer programming, Object-Oriented Programming, Numerical Analysis, Algorithms & Data Structures
- **Data Management:** Data Structures, Database Management (SQL, NoSQL, VectorDB)
- **Machine Learning:** Advanced Python Programming, Linear Algebra and Statistics
- **Microprocessor Related:** Microcontroller and Microprocessor, VLSI Technology.

Extracurricular Activities

2022 **Coding Problem Setter**, 2022 ETE DecaHertz Programming Contest

Bangladesh

2023 **Organizer & Program Co-ordinator**, Infixon

Bangladesh

2023 **Organizer**, Tarunner Uthsob by CUET Debating Society

Bangladesh

Leadership Experiences and Affiliations

CUET Debating Society

Chittagong, Bangladesh

VICE-PRESIDENT

July. 2023 - Sept. 2024

- Arranged a National Level Inter-University Debate Competition and Cultural Program.
- Participants at Inter University English Debate Competition.

House of the Language and Literature, CUET

Chittagong, Bangladesh

ORGANIZING SECRETARY

August. 2023 - Sept. 2024

- Arranged a literature fest for all the members of the university to explore books.
- Arranged weekly book reading sessions with fellow book lovers.